

# Andresa Rodrigues de Campos

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Amazon  
Applied Scientist  
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Education     **Carnegie Mellon University, Pittsburgh, PA USA**

Ph.D., Physics, 2023 – Advisor: Scott Dodelson, Ph.D.

**São Paulo State University, SP Brazil**

M.S., Physics, 2018 – Advisor: Rogerio Rosenfeld, Ph.D.

**Universidade Federal de São João del-Rei, MG Brazil**

B.S., Physics (GPA: 9.23/10), Minor in Computational Physics, 2015

Dissertation     *Dark Energy Science from 100 Million Galaxies: AI-Driven Analysis and Data-Intensive Techniques for Cosmological Discovery*

Research     **Amazon**

Applied Scientist

November 2024 – present

- Designed and deployed a novel lossless prompt compression system for LLM-powered process mining, achieving 40% input token reduction and 30% latency improvement across 1.2M production events. Discovered that LLMs can learn compression dictionaries through in-context learning.
- Developed and deployed an ML classification system for property tax asset operations, improving accuracy from 64% to 95% and enabling automated classification for 92.6% of 180K+ annual asset additions.
- Built an AI-driven contract classification system using LLMs and chain-of-thought prompt engineering, achieving 88% accuracy across 48K contracts and increasing tax automation coverage from 35% to 95%+.
- Designed and deployed RAG pipelines for contract term extraction across multiple business domains, reducing manual review by 300+ hours/year with 91.6% accuracy.
- Designing a two-phase clustering algorithm combining cosine similarity with LLM validation for automated customer onboarding at scale. In progress.

## **Dark Energy Survey Collaboration**

Analysis Team Lead

October 2019 – December 2023

- Developed a machine learning pipeline for supercomputer analysis, enhancing galaxy distance estimation from photometry in the largest galaxy catalog to date (100M galaxies), to improve understanding of dark energy.
- Employed self-organizing maps and Bayesian statistics to reduce galaxy distance estimation uncertainty by 66%, satisfying Department of Energy goals.
- Led a team of 5 scientists conducting hypothesis testing, consistently delivering high-quality results on schedule.

## **Carnegie Mellon University**

Doctoral Researcher

September 2018 – December 2023

- Developed a statistical method for empirical model selection to address limitations in current modeling methods.
- Conducted comprehensive analysis identifying a 30% likelihood of inadequacy in a prevalent astrophysical model.
- Induced a paradigm shift in astrophysics, leading to method's adoption by leading research collaborations (Hyper Suprime-Cam Subaru Telescope, Dark Energy Survey).

## Teaching **Teaching Assistant – Carnegie Mellon University**

Physics II for Engineering and Physics Students, Spring 2019 & Fall 2018

Stars, Galaxies and the Universe, Fall 2018

## **Lecturer – Universidade Federal de São João del-Rei**

Fundamentals of Physics I, Fall 2015

Introductory Experimental Physics, Fall 2015

## **Course Tutor – Universidade Federal de São João del-Rei**

Calculus II, Fall 2012

Treatment and Representation of Exp. Measurements, Spring 2012 & Fall 2011

Publications Author or co-author of 79 peer-reviewed papers and preprints.

Total citations: 8,943; h-index: 41.

Profiles: ORCID | ADS Library | Google Scholar

### *First-author papers:*

- **Andresa Rodrigues de Campos**, David Lee, Imry Kissos, Piyush Paritosh, “Loss-less Prompt Compression via Dictionary-Encoding and In-Context Learning: Enabling Cost-Effective LLM Analysis of Repetitive Data.” arXiv:2604.13066v1 (2026)
- **Campos, A.**; Yin, B.; Dodelson, S.; Amon, A.; Alarcon, A.; Sánchez, C.; Bernstein, G. M.; Giannini, G.; Myles, J.; Samuroff, S.; et al., “Enhancing weak lensing redshift distribution characterization by optimizing the Dark Energy Survey Self-Organizing Map Photo-z method.” arXiv:2408.00922 (2024).
- **Campos, Andresa**; Samuroff, Simon; Mandelbaum, Rachel, “An empirical approach to model selection: weak lensing and intrinsic alignments.” MNRAS, arXiv:2211.02800

(2023).

- Lemos, P.; Raveri, M.; **Campos, A.**; Park, Y.; Chang, C.; Weaverdyck, N.; Huterer, D.; Liddle, A. R.; Blazek, J.; Cawthon, R.; et al., “Assessing tension metrics with dark energy survey and Planck data.” MNRAS, arXiv:2012.09554 (2021) – Equal contributions from the first three authors.

*Key co-authored papers (by date, newest first):*

- **DES Collaboration**; Abbott, T. M. C.; Adamow, M.; et al., “Dark Energy Survey Year 6 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing.” arXiv:2601.14559 (2026).
- **DES Collaboration**; Abbott, T. M. C.; Aguena, M.; et al., “Dark Energy Survey Year 6 Results: Cosmological Constraints from Cosmic Shear.” arXiv:2602.10065 (2026).
- Yin, B.; Amon, A.; **Campos, A.**; et al., “Dark Energy Survey Year 6 Results: Redshift Calibration of the Weak Lensing Source Galaxies.” arXiv:2510.23566 (2025).
- Samuroff, S.; **Campos, A.**; Porredon, A.; Blazek, J., “Joint constraints from cosmic shear, galaxy-galaxy lensing and galaxy clustering: internal tension as an indicator of intrinsic alignment modelling error.” Open J. Astrophys., arXiv:2402.15573 (2024).
- Samuroff, S.; Mandelbaum, R.; Blazek, J.; **Campos, A.**; MacCrann, N.; Zacharegkas, G.; Amon, A.; Prat, J.; Singh, S.; Elvin-Poole, J.; et al., “The Dark Energy Survey Year 3 and eBOSS: constraining galaxy intrinsic alignments across luminosity and colour space.” MNRAS, arXiv:2212.11319 (2023).
- Lemos, P.; Weaverdyck, N.; Rollins, R. P.; Muir, J.; Ferté, A.; Liddle, A. R.; **Campos, A.**; Huterer, D.; Raveri, M.; Zuntz, J.; et al., “Robust sampling for weak lensing and clustering analyses with the Dark Energy Survey.” MNRAS, arXiv:2202.08233 (2023).
- Myles, J.; Gruen, D.; Amon, A.; Alarcon, A.; DeRose, J.; Everett, S.; Dodelson, S.; Bernstein, G. M.; **Campos, A.**; Harrison, I.; et al., “Mapping variations of redshift distributions with probability integral transforms.” MNRAS, arXiv:2210.03130 (2022).
- Secco, L. F.; Samuroff, S.; Krause, E.; Jain, B.; Blazek, J.; Raveri, M.; **Campos, A.**; Amon, A.; Chen, A.; Doux, C.; et al., “Dark Energy Survey Year 3 results: Cosmology from cosmic shear and robustness to modeling uncertainty.” Phys. Rev. D, arXiv:2105.13544 (2022).
- Breivik, K.; Connolly, A. J.; Ford, K. E. S.; Jurić, M.; Mandelbaum, R.; Miller, A. A.; Norman, D.; Olsen, K.; O’Mullane, W.; Price-Whelan, A.; et al., “From Data to Software to Science with the Rubin Observatory LSST.” arXiv:2208.02781 (2022).
- Myles, J.; Alarcon, A.; Amon, A.; Sánchez, C.; Everett, S.; DeRose, J.; McCullough, J.; Gruen, D.; Bernstein, G. M.; Troxel, M. A.; et al., “Dark Energy Survey Year 3 results: redshift calibration of the weak lensing source galaxies.” MNRAS, arXiv:2012.08566 (2021).
- Doux, C.; Baxter, E.; Lemos, P.; Chang, C.; Alarcon, A.; Amon, A.; **Campos, A.**; Choi, A.; Gatti, M.; Gruen, D.; et al., “Dark energy survey internal consistency tests of the joint cosmological probes analysis with posterior predictive distributions.” MNRAS, arXiv:2011.03410 (2021).

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| Invited Talks                | <b>From DES Y3 to Y6: SOMPZ's Journey Towards Better Redshifts</b> ,<br>Urbana-Champaign US, October 2023<br><b>Weak Lensing Photo-Z for DES Y6</b> , Cambridge UK, July 2022<br><b>Self-Organizing Maps</b> , Duke NC, May 2022<br><b>SOMPZ for Lens and Source Galaxy Samples</b> , Duke NC, May 2022<br><b>Empirical Model Selection</b> , Duke NC, May 2022<br><b>Tension Metrics</b> , Sussex UK, November 2019                                                                                                                                                                                                                                                                                                                                  |
| Awards and                   | <b>Best Poster Award</b> , Amazon Internal Science Fair, 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Fellowships                  | <b>Data Science for All Women</b><br>Data science fellowship by Correlation One – Summer 2023<br><b>Dark Energy Survey Builder</b><br>Two years of full-time equivalent work on infrastructure – Fall 2022<br><b>Dark Energy Survey Early Career Scientist</b><br>Grant for participating in the Fall 2019 DES Collaboration Meeting<br><b>Dark Energy Survey Observing Award</b><br>Grant for observing for DES using the DECam at Cerro Tololo in Chile<br>7 nights in January 2019 – DES Year 6<br><b>CNPq Scholarship, São Paulo State University, SP, Brazil</b><br>Graduate Research Fellowship, awarded in 2016 (2 years duration)<br><b>FAPEMG Scholarship, UFSJ, MG, Brazil</b><br>Undergraduate Research Fellowship, awarded in 2013 & 2014 |
| Service to<br>the Profession | <b>Science Paper Reviewer, Amazon Internal Conferences</b><br>Reviewed 9+ papers across AMLC 2025, CSS 2026, and AMLC 2026, evaluating research for scientific rigor, novelty, and practical impact. Recognized with a reviewer icon for AMLC 2025.<br><b>McWilliams Software Development Series</b><br>Co-organizer: Sessions with tutorials and talks related to software development and computing resources, 2021 – 2023                                                                                                                                                                                                                                                                                                                          |
| Students and<br>Mentees      | <b>Boyan Yin</b> : Undergraduate, Carnegie Mellon University<br><b>Hannah Varekamp</b> : Undergraduate, Carnegie Mellon University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Leadership                   | <b>Dark Energy Survey</b><br>Analysis Team Lead, Weak Lensing Redshift Distributions, 2020 – 2023<br>Analysis Team Co-Lead, Samplers and Tensions, 2018 – 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Collaborations               | <b>The Dark Energy Survey (DES)</b><br><b>The LSST Dark Energy Science Collaboration (DESC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Grants                       | <b>Extracting Information from the Large Scale Structure of the Universe</b><br>XSEDE Allocation: AST200006, 2021 (200,000 CPU hours)<br>Co-lead and authored the proposal. PI: Scott Dodelson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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| Languages and Skills    | <b>Languages:</b> English (fluent), Portuguese (native), Spanish (advanced)<br><b>Programming Languages:</b> Python (7+ yoe), Java, R, Bash, FORTRAN<br><b>Machine Learning:</b> PyTorch, Scikit-Learn, AutoGluon, TensorFlow, Keras<br><b>NLP &amp; LLMs:</b> Prompt Engineering, RAG, Chain-of-Thought, In-Context Learning<br><b>Data Analysis:</b> Jupyter, NumPy, CuPy, SciPy, Pandas<br><b>Data Visualization:</b> Matplotlib, Seaborn, Plotly, Altair<br><b>Markup Languages:</b> L <sup>A</sup> T <sub>E</sub> X, basic knowledge of HTML + CSS<br><b>Databases:</b> SQL, PostgreSQL<br><b>High Performance Computing:</b> Spark, PySpark, SLURM, MPI, mpi4py<br><b>Tools:</b> Git, GitHub, Linux |
| Outreach and Leadership | <b>President – Tartan Salsa</b><br>Student organization at Carnegie Mellon University promoting Latin American dances and culture. Led weekly classes averaging 80 attendees, grew membership by 70%, and secured over \$20,000 in funding over two years. 2021 – 2023                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Relevant Coursework     | <b>Introduction to Machine Learning</b><br><b>Neural Networks and Deep Learning in Science</b><br><b>Modern Programming for Data Scientists</b><br><b>Introduction to Large-Scale Computing in Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |